

Chakib OUZANE

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Education

MSc	Master's Degree in Computer Science ESI SBA (l'École Supérieure d'Informatique - Sidi Bel Abbès) Specialty: Computer Systems Engineering (ISI)	2020 – 2025 Sidi Bel Abbès, Algeria
Eng	State Engineer Degree in Computer Science ESI SBA (l'École Supérieure d'Informatique - Sidi Bel Abbès) Specialty: Computer Systems Engineering (ISI)	2020 – 2025 Sidi Bel Abbès, Algeria
BAC	Baccalaureate Diploma Branch: Mathematics	09/2020 Sidi Bel Abbès, Algeria

Experiences

Internship at CESI LINEACT Research Lab , Machine learning & Network security	04/2025 - 09/2025 Aix-en-Provence, France
- Comprehensive literature review of Researches on security in IoT Networks. - Designing, Implementing and Simulating Flooding Attacks on RPL networks. - Comprehensive literature review of Researches on Machine Learning countermeasures for IoT Security. - Proposing a FL based framework for detecting routing attacks in IoT environments that rely on RPL.	
Internship at GSH (Groupe Sociétés HASNAOUI) , Network security	06/2023 - 09/2023 Sidi Bel Abbès, Algeria
- Experimentation with the SOPHOS software. - Setting SOPHOS in virtual environments, testing filtering policies, Intrusion Prevention (IPS), and deep packet inspection for network traffic protection.	

Publications

DRIFT: Distributed RPL Injection Flooding Threat in IoT Networks	08/06/2026 London, United Kingdom
- Publication at the 14th International Conference on Frontiers of Intelligent Computing: Theory and Applications (FICTA 2026). - Conference paper status: Accepted — Best Paper Award	
TIDE: Threat of Internal DIS Exploit in RPL-based IoT Networks	2027 Washington DC, USA
- Targeting: IEEE International Conference on Communications (ICC), 2027. - Conference paper status: In Preparation	

Projects

Machine Learning-Based Spam Detection in Chicago Hotel and Restaurant Reviews	12/2024
- Built a machine learning pipeline to classify reviews as spam or non-spam using text and metadata features. - Performed data cleaning, feature engineering (TF-IDF, sentiment scores, reviewer behavior), and model evaluation (Logistic Regression, Random Forest, XGBoost). - Achieved 89.19% test accuracy with XGBoost; analyzed class imbalance impact and proposed improvements for F1 score.	

Machine Learning Models for Customer Analytics

- Applied K-means clustering to segment customers by type, gender, and payment method, providing actionable insights for targeted marketing.
- Developed a K-Nearest Neighbors (K-NN) classifier on the Bank Marketing dataset to predict customer responses using supervised learning.
- Used R and ML libraries for data preprocessing, model training, and result interpretation to support decision-making.

Supervised Machine Learning for Sales Data Analysis and Prediction

- Performed data preprocessing and feature selection on a supermarket sales dataset.
- Visualized sales trends and spending patterns across branches, customer types, and products using R and ggplot2.
- Prepared dataset splits for training and testing supervised ML models.

Jeunesse Volontaire

2024

- A Website + Platform built for an environmental organization.
- Technologies used: SvelteKit & Svelte, Supabase, Sanity CMS, Zod, Tailwind.

jeunessevolontairesba.dz

Alphabit Club Official Website

2023

- I worked with a team to develop a website for the AlphaBit Club, main goal was to attract companies to finance the club events.
- Technologies used: React, Tailwind CSS, Headless UI, Firebase, Axios.

www.alphabit.club

Innovium

03/2023 - 06/2023

- A web application to digitize the application process for startups and final-year projects, developed as part of my 3rd year in a team.
- Technologies used: Svelte TypeScript, Tailwind CSS, DaisyUI, svelte-i18n.

github.com/boite-informatique/startup-front

ESI Hub

03/2022 - 06/2022

- Contributed to the backend development of a social network platform for students at ESI SBA, developed as part of my 2nd year in a team.
- Technologies used: Node.js, Express, MongoDB, Mongoose, JWT, Multer, Socket.io.

github.com/boite-informatique/esi-hub-backend

Skills

Languages: C, Python, R, Java, JavaScript.

Web Development: Svelte & SvelteKit, React, Node.js & Express.js, REST APIs & GraphQL.

Databases: Supabase, MongoDB, PostgreSQL, Redis.

Networking: IoT, Routing Protocols (RPL, EIGRP, OSPF, BGP).

AI & ML: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib.

Mathematics: Numerical Linear Algebra, Probability & Statistics, Numerical Methods, Graph Theory.

Security: Networking Attacks, Cryptography Algorithms, Blockchain, IDS/IPS, Firewalls, VPNs, DMZ.

Tools: Git & GitHub, Docker, Contiki-NG/Cooja & GNS3, LaTeX.

Languages

English
Fluent

French
Advanced level - TCF C1

Arabic
Native language

German
Goethe Certificate B1

Volunteering

Alphabit Club - Technical Team Member

Member of the technical team responsible for creating and maintaining the Alphabit website, authoring hackathon challenges, and performing other technical tasks during events.

2022 - 2024

Algeria

alphabit.club 

Internationale Jugendgemeinschaftsdienste (ijgd), Germany

Completed ijgd training on youth work and group leadership, allowing me to lead international workcamps in Germany and other EU countries.

02/2024

Hannover, Germany

www.ijgd.de 

EYE 2025 (European Youth Event)

Participation in the European Youth Event organized by European Parliament.

06/2025

Strasbourg, France

Jeunesse Volontaire

A local environmental and social action organization in Algeria.

2016 - 2024

Sidi Bel Abbes, Algeria

IJGD (08/2023 - 09/2023)

A volunteer workcamp in Germany, to assist with the showcasing of art exhibitions.

09/2023

Neunkirchen, Germany

IJGD (08/2019 - 09/2019)

A volunteer workcamp in Bärnau-Tachov, for the maintenance of a local historical park.

09/2019

Bärnau, Germany

IJGD - Camp for Future

A volunteer workcamp focused on anti-coal activism, organized at the Hambach Forest.

08/2018

Hambach, Germany

Interests

Cybersecurity, Internet Freedom, FOSS (Free Open Source Software), Web3, Football, Geography, Urban planning, Travel and discovering other cultures.